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CHECKLIST

SMALL CRAFT - WASTE SYSTEMS - PART 1: WASTE WATER RETENTION

Ref.: EN ISO 8099-1:2018 (ISO 8099-1:2018)

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	

	Subject to check	Clause	Requirements	Checked ?
1	Any toilet in a retention system is connected solely to a holding tank.	4.1	[Yes ?]	
2	There shall be no Y-valve placed between the toilet and the holding tank.	4.1	[Yes ?]	
3	Craft with permanently installed holding tanks are fitted with a discharge connection as specified in Annex A.	4.2	[Yes ?]	
4	Any through-hull fittings for sewage are fitted with valves which are capable of being secured in the closed position.	4.2	[Yes / NA ?]	
5	Connecting hoses and pipings are securely fastened in position to prevent damage by abrasion or vibration.	6.1	[Yes ?]	
6	Piping or hose between the toilet and holding tank, and between the tank and the pump-out fitting, are as short as practicable.	6.1	[Yes ?]	
7	Retention systems with the possibility of overboard discharge are fitted with a seacock at the through-hull fitting.	6.2	[Yes / NA ?]	
8	Any seacock used for overboard discharge in accordance with ISO 9093:2021 are capable of being secured in the closed position.	6.2	[Yes / NA ?]	
9	**Signage:** If a manual relief valve is fittet for the tank venting a sign is installed in the vicinity of the pump out fitting. Symbols or working in language acceptable in the country of use to indicate that the relief valve must be opened prior to pump out.	7.2.2	[Yes / NA ?]	
10	The holding tank is securely fastened and located independently of any connecting piping.	8.1.1	[Yes ?]	
11	The minimum level of holding-tank content is observable when the holding tank is 3/4 full by volume, when the tank is viewed in a readily accessible location, or indicated by other means.	8.1.2	[Yes ?]	
12	Holding-tank fittings and connections are accessible for inspection and maintenance.	8.1.3	[Yes ?]	
13	Holding tanks of capacity > 40 l have an accessible sealable (i.e. gastight and watertight) minimum opening of 75 mm diameter for flushing, cleaning and maintenance.	8.1.4	[Yes / NA ?]	
14	Holding tanks does not have common walls, tops or bottoms with fuel or portable-water tanks.	8.1.5	[Yes ?]	
15	If portable holding tank(s) on the craft, it is not connected to any through-hull fitting.	8.3.1	[Yes / NA ?]	
16	If portable toilets are equipped with a discharge fitting the tank shall be considered as a permanently installed holding tank.	8.3.1	[Yes / NA ?]	
17	**Label:** How to diconnect, transport and empty portable holding tank(s). **Affixed:** on the tank.	8.3.5	[Yes / NA ?]	
18	Dimension of pump-out fitting as shown in figure A.1.	9.1	[Yes ?]	

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X

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19	**Label:** Pump-out fitting identified by marking, with at least the symbol:  **Affixed:** on the fitting or in its vicinity.	9.3	[Yes ?]	
20	The fittings have a sealing cap.	9.4	[Yes ?]	
21	If a cap retention system is used, it not impede the proper function of the pump-out.	9.4	[Yes / NA ?]	
22	Pump-out fittings are readily accessible with access for pump-out connections.	9.5	[Yes ?]	
23	Pump-out fittings are located as far as practicable from the fuel tank fill and potable water fittings.	9.5	[Yes ?]	
24	**Label:** Prefabricated holding tanks legibly marked with: -name or trademark of the manufacturer; -name/model number of the system; -symbol (see 9.3) or text "Toilet waste tank" in language acceptable in the country of use; -tank capacity (litres). **Affixed:** on the holding tank.	10	[Yes ?]	

The following questions shall be filled in by the watercraft manufacturer and appropriate documentation shall be submitted to the inspector for verification.

	Subject to check	Clause	Requirements	Checked ?
25	The system prevent the emission of vapour and liquids within the craft.	4.3	[Yes ?]	
26	The system is capable of operation throughout an ambient temperature range of +1 °C to +60 °C and withstand, when empty, an ambient temperature range of –40 °C to +60 °C.	4.4	[Yes ?]	
27	The system is capable of operation, i.e. discharge of sewage from the toilet to the retention system, when the boat is heeled at all angles up to 20 ° for monohull sailing craft and 7 ° for other craft.	4.5	[Yes ?]	
28	Back siphoning is prevented from raw water intakes and discharge outlets up to a heel angle to either side of at least 30° for monohull sailing craft, 20° for other craft and a trimmed condition at the bow or stern of at least 10°.	4.6	[Yes ?]	
29	Back siphoning of the contents and escape of gas from the holding tank back through the toilet fixture shall be prevented when the boat is heeled at all angles up to 30° for monohull sailing craft, 20° for other craft and a trimmed condition at the bow or stern of at least 10°.	4.7	[Yes ?]	
30	Escape of sewage from the holding tank to the exterior of the craft shall be prevented when the boat is heeled at all angles up to 30° for monohull sailing craft, 20° for other craft, at 90 % of tank capacity and to the interior of the craft under maximum anticipated conditions of heel or trim, i.e. 45° for monohull sailing craft, 30° for other craft.	4.8	[Yes ?]	
31	If permanently installed the retention system including all tanks, connecting piping, hoses, and fittings, is tested to withstand a pressure of 20 kPa for a period of 5 min without leaking.	4.10	[Yes ?]	

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X

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32	The tank withstand a negative pressure of 20 kPa for a period of 5 min without permanent deformation.	4.10	[Yes ?]	
33	Materials are resistant to the effects, listed in Clause 5.	5	[Yes ?]	
34	Hoses and piping suitable for use in sewage systems.	6.1	[Yes ?]	
35	Piping or hose between the toilet and holding tank, and between the tank and the pump-out fitting inner surface be smooth and without convolutions to permit free flow of sewage.	6.1	[Yes ?]	
36	Piping or hose between the toilet and holding tank, and between the tank and the pump-out fitting inner surface have an inside diameter in conformity with the toilet manufacturer's recommendations, or have a minimum inside diameter of 38 mm, if no recommendations are provided.	6.1	[Yes ?]	
37	The system provide for venting of gases within the system to the exterior of the craft at heel angles up to 20° at 90 % of tank capacity.	7.1.1	[Yes ?]	
38	The inside diameter of fittings to which vent piping is connected is not be less than 75 % of the inside diameter of the piping.	7.1.2	[Yes ?]	
39	The design and construction of the vent system minimize clogging due to the contents of the tank or as a result of weather conditions.	7.1.3	[Yes ?]	
40	The vent is capable of resisting, without damage, a negative pressure of 20 kPa.	7.1.3	[Yes ?]	
41	The minimum flow area through vent screens and equivalent flow resistance of any filters installed in the vent system are not be less than the smallest flow area in either the vent pipe or its fittings.	7.1.4	[Yes ?]	
42	Vent pipe complies with standard and documentation is available by the manufacturer.	7.2	[Yes ?]	
43	If rigid tanks with capacity of less tan 400 l are used, the minimum inside diameter of the vent pipe amount 19 mm.	7.2.1	[Yes / NA ?]	
44	If rigid tanks with capacity of less than 400 l and the vent pipe inside diameter of not less than 16 mm are used, the tank is fitted with an automatic (vacuum operated) or manual relief valve with a minimum combined area of 1 100 mm2.	7.2.1	[Yes / NA ?]	
45	If rigid tanks with capacity of 400 l and greater are used, the minimum inside diameter of the vent pipe amount 38 mm.	7.2.2	[Yes / NA ?]	
46	If rigid tanks with capacity of 400 l and greater and multiple vent pipes with at least 19 mm diameter are used the combined cross sectional flow area is at least equivalent to that of a single vent pipe with an area of 1100mm2.	7.2.2	[Yes / NA ?]	
47	If rigid tanks with capacity of more than 400 l and vent pipe inside diameter of not less than 16 mm are used, the tank is fitted with an automatic (vacuum operated) or manual relief valve with a minimum combined area of 1100 mm2.	7.2.2	[Yes / NA ?]	
48	Flexible (collapsible) tanks have at least one vent of inside diameter minimum 16 mm.	7.2.3	[Yes / NA ?]	
49	The permanently installed holding tank provide removal of at least 90 % of its contents through the pump out fitting.	8.2.1	[Yes / NA ?]	
50	Baffles in permanently installed holding tanks have openings to allow sewage and vapour to flow freely across the top and bottom.	8.2.2	[Yes / NA ?]	
51	Fittings of permanently installed holding tanks, including the covers of clean-out openings, are designed and constructed to ensure a gastight and watertight closure.	8.2.3	[Yes / NA ?]	

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X

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52	The internal diameter of the vent line for portable holding tanks is not less than 16mm.	8.3.2	[Yes / NA ?]	
53	Portable holding tank installed with quick disconnect at the tank opening with a closing device permanently attached to the tank, to ensures a watertight seal during transport of the tank.	8.3.2	[Yes / NA ?]	
54	All portable holding-tank openings are sealed with watertight and gastight closing devices.	8.3.3	[Yes / NA ?]	
55	Threads are in accordance with ISO 228-1.	9.2	[Yes ?]	

	Instructions/Warnings to be included in the owner's manual			
57	Operating and maintanance.	11	[Yes ?]	
58	Y-valve use (securing, avoidance of inadvertend discharge).	11	[Yes / NA ?]	
59	Capacity of the holding tank [litres].	11	[Yes ?]	
60	Chemical acceptance for use; cleaning materials;additives;anti-freeze solutions.	11	[Yes ?]	
61	Pump-out procedure.	11	[Yes ?]	
62	Instruction that the system shall be empty during storage at freezing temperatures.	11	[Yes ?]	

Comments:

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X